Rebound: Centralised Game Logging System for Athlete Performance Analysis

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***Abstract*—Systematic performance analysis in sports relies on the availability of structured and accessible performance data. While professional sports organizations employ sophisticated analytics platforms to capture and interpret athlete statistics, comparable tools for amateur and grassroots athletes remain limited. As a result, performance records at these levels are often fragmented, manually maintained, or entirely undocumented, restricting opportunities for longitudinal analysis and informed decision-making. This study presents the design and evaluation of a centralized game logging system that enables structured recording and analysis of athlete performance data. The proposed system provides a unified platform for documenting game events, capturing player statistics, and generating analytical insights from accumulated performance records. By integrating a mobile-based interface with a cloud-hosted data infrastructure, the system organizes game logs and player statistics into a standardized data model that supports consistent performance tracking over time. The evaluation demonstrates how centralized data collection improves record accessibility and enables meaningful performance analysis through aggregated metrics and visual trends. The findings highlight the potential of structured game logging systems to enhance performance documentation and analytical capabilities for athletes and coaches operating outside professional sports environments.**

***Index Terms*—Sports analytics; athlete performance tracking; game logging systems; sports data management; amateur sports technology**

I. Introduction

Data-driven decision-making has become an essential component of modern sports performance analysis. Professional sports organizations increasingly rely on analytical platforms to monitor athlete performance, evaluate strategies, and guide long-term player development. Through structured performance datasets and analytical tools, teams can generate insights that support training optimization, match preparation, and talent identification. However, access to such sophisticated performance tracking systems remains largely limited to elite sports environments.

In contrast, athletes participating in amateur or semi-professional sports frequently lack accessible tools to systematically document their sporting activities. Records of matches played, player statistics, and game outcomes are often maintained informally or remain undocumented. The absence of structured performance records restricts athletes and coaches from analyzing performance trends, monitoring improvement over time, or maintaining a comprehensive history of athletic participation. As a result, meaningful performance evaluation and long-term development become difficult to achieve in non-professional sports contexts.

A. Background and Motivation

Recent advancements in digital platforms have demonstrated the potential of centralized data systems to support structured sports data management. By enabling athletes and coaches to record match statistics, player contributions, and game outcomes within a unified system, these platforms can transform fragmented records into organized datasets suitable for performance analysis. Centralized game logging allows athletes to maintain detailed histories of matches played, teams involved, scores, and statistical contributions, creating a structured foundation for analytical evaluation.

Additionally, analytical tools integrated within such platforms can generate visual insights from recorded data. Performance metrics such as average points per game, rebounds, assists, and shooting efficiency can be analyzed through graphs, timelines, and statistical summaries. These visual representations help translate raw performance data into interpretable insights that assist athletes and coaches in evaluating strengths,

identifying areas for improvement, and making informed training decisions.

B. Problem Statement

Despite the increasing importance of sports analytics, several challenges remain for athletes operating outside professional sports ecosystems:

- Lack of centralized platforms for recording and organizing game statistics.
- Limited access to analytical tools for evaluating individual and team performance.
- Absence of structured digital portfolios that document an athlete's sporting journey.

Existing sports analytics platforms primarily focus on professional leagues or specialized teams, leaving amateur athletes with limited options for systematic performance documentation and analysis.

C. Research Gap

Current solutions in sports performance tracking typically address isolated aspects of athlete management, such as match statistics recording or social interaction features. However, few platforms integrate:

- Structured game logging and player statistics management,
- Performance analytics and visualization,
- Athlete portfolio generation, and
- Team-level performance analysis

within a single unified platform designed for accessibility and usability.

This gap highlights the need for a centralized system capable of collecting, organizing, and analyzing sports performance data while simultaneously enabling athletes to maintain a documented record of their sporting activities.

D. Proposed Approach

To address these limitations, this work presents *Athlinix*, a centralized sports performance tracking and analytics platform designed for athletes and coaches. The proposed system enables structured recording of game events, player statistics, and match outcomes while generating analytical insights from the collected data. The platform integrates game logging, performance visualization, team analytics, and athlete profile management within a unified digital environment.

Through centralized data management and analytical tools, Athlinix allows athletes to monitor performance trends, maintain detailed match histories, and build comprehensive digital portfolios representing their sports journeys.

E. Contributions

The primary contributions of this work are:

- Design of a centralized sports data management platform for recording games and player statistics.
- Integration of performance analytics tools that generate visual insights from recorded match data.
- Development of digital athlete portfolios that document sporting activities and achievements.
- Support for both individual and team-level performance evaluation within a unified system.

F. Paper Organization

The remainder of this paper is organized as follows. Section II reviews related work in sports analytics and performance tracking systems. Section III presents the system architecture and database design. Section IV describes the implementation methodology and experimental setup. Section V discusses the results and analytical outcomes generated by the platform. Finally, Section VI concludes the study and outlines potential future enhancements.

II. Related Work

The growing importance of data analytics in sports has led to the development of numerous technologies and platforms designed to capture and analyze athlete performance. Professional sports organizations widely employ advanced analytical systems to collect large volumes of performance data and derive insights that support strategic decision-making. Research in sports analytics has explored various methods of performance evaluation, including statistical modeling, player tracking technologies, and data-driven training optimization. These approaches have demonstrated the value of structured data in understanding athlete performance and improving competitive outcomes.

Several commercial platforms have also emerged to facilitate sports performance tracking and analysis. Applications such as *Strava* primarily focus on endurance sports, providing users with the ability to record activities, monitor fitness metrics, and analyze performance trends over time. Similarly, wearable-based systems and mobile fitness trackers enable athletes to monitor physiological parameters such as heart rate, distance covered, and training intensity. While these platforms provide valuable insights for individual fitness monitoring, they are largely designed for activities such as running or cycling and do not comprehensively support the recording of structured game events in team sports.

Other platforms such as *Hudl* and *GameChanger* provide tools for teams to record match statistics and analyze gameplay. Hudl is widely used for video analysis and performance review, enabling teams to study game footage and identify tactical patterns. GameChanger offers scorekeeping and basic statistical tracking for organized sports teams. However, these systems are often designed for structured leagues or institutional environments where dedicated infrastructure and resources are available. Additionally, many of these platforms emphasize team-level management and video analysis rather than maintaining long-term, structured performance records for individual athletes.

Despite the availability of these technologies, a gap remains in providing accessible and structured systems that enable athletes to record and analyze their performance across multiple games. Many existing solutions either focus on fitness

tracking, professional-level analytics, or team administration, leaving limited support for centralized documentation of individual game statistics and long-term performance histories. As a result, athletes participating in amateur team sports frequently lack tools that allow them to systematically record game events and analyze performance trends over time.

The system proposed in this study addresses this gap by introducing a centralized game logging platform designed specifically to support structured recording and analysis of athlete performance data. By combining game logging, statistical tracking, performance analytics, and athlete portfolio generation within a unified system, the proposed approach aims to provide a more accessible framework for documenting and analyzing sports performance in amateur environments.

III. System Architecture

The platform is designed to allow athletes and coaches to record detailed information about games and player performance through a mobile application. Users can create teams, log matches, record player statistics, and view analytical summaries derived from the stored data. The backend infrastructure maintains a centralized repository that organizes athlete profiles, team information, game records, and statistical metrics.

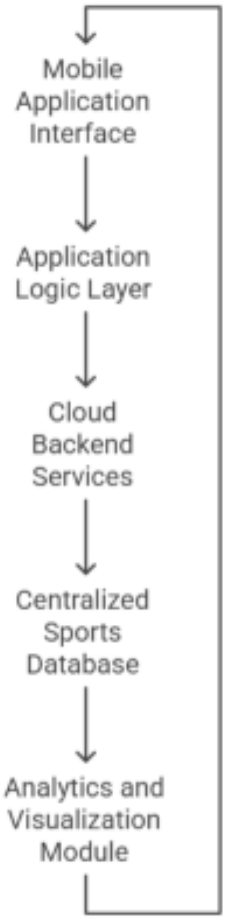


Fig. 1. System Architecture Workflow

This architecture ensures that performance data entered through the mobile interface is securely stored in a centralized database and processed to generate analytical insights.

A. User Interface Layer

The user interface layer provides athletes and coaches with access to the system through a mobile application. The interface enables users to create and manage profiles, record game information, input player statistics, and view analytical dashboards. The design prioritizes accessibility and ease of use so that athletes can quickly log game events and review performance data without requiring specialized technical knowledge.

Through the interface, users can access features such as game logging, team management, performance visualization, and athlete profile management. The interface also allows users to view performance metrics through graphs and statistical summaries.

B. Application Logic Layer

The application logic layer manages the core functionality of the platform. It processes user inputs from the interface, validates data, and coordinates interactions between the application and the backend services. This layer handles operations such as creating game records, updating player statistics, retrieving historical performance data, and generating statistical summaries.

The logic layer ensures that data entered by users is structured according to predefined schemas, enabling consistent storage and analysis of performance records.

C. Data Management Layer

The data management layer is responsible for storing and organizing all information related to athletes, teams, games, and player statistics. A centralized cloud database maintains structured records that allow performance data to be stored across multiple games and seasons.

The database includes entities such as user profiles, teams, game logs, and player performance statistics. Each game record contains details such as participating teams, match results, and individual player contributions. Player statistics are linked to specific games, enabling longitudinal tracking of performance across multiple matches.

By maintaining structured records within a centralized database, the system supports efficient retrieval of historical data and enables the generation of analytical insights based on accumulated statistics.

D. Analytics and Visualization Module

The analytics and visualization module is responsible for transforming recorded game statistics into interpretable performance insights for both athletes and teams. The module processes structured game logs stored in the centralized database and applies statistical computations to generate key performance indicators (KPIs). These metrics are then presented through analytical dashboards, graphs, and performance trend visualizations within the application interface.

The analytical framework relies on aggregated statistical measures derived from player game records and team match outcomes. The following formulations describe the mathematical metrics used by the system to evaluate performance consistency, player contribution, and team effectiveness across multiple games.

1) Athlete Performance Metrics: Athlete-level analytics focus on evaluating individual performance trends across multiple games using statistical measures such as averages, variability, and composite performance indicators.

The average performance metric across multiple games is calculated using the arithmetic mean:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{1}$$

where x_i represents the recorded statistic (such as points, rebounds, or assists) in game i, and n represents the total number of games played.

To evaluate the consistency of performance, the standard deviation is computed as:

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{2}$$

To quantify stability relative to average performance, a consistency ratio is defined as:

$$CR = \frac{\sigma}{\bar{x}} \tag{3}$$

In addition to statistical measures, the system computes a player impact score representing the overall contribution of an athlete in a given match:

$$ImpactScore = Points + Rebounds + Assists + Steals - Fouls \tag{4}$$

The average impact score across multiple matches is calculated as:

$$AvgImpact = \frac{1}{n}\sum_{i=1}^{n} ImpactScore_i \tag{5}$$

These athlete-level metrics enable the system to generate performance trend graphs, identify standout individual performances, and evaluate long-term player development.

2) Team Performance Metrics: Team-level analytics evaluate collective performance using aggregated scoring statistics and match outcomes.

The total points scored by a team across all matches are computed as:

$$Points_{scored} = \sum_{i=1}^{n} P_i \tag{6}$$

Similarly, the total points conceded are calculated as:

$$Points_{conceded} = \sum_{i=1}^{n} O_i \tag{7}$$

The average points scored per game are calculated as:

$$AvgPointsFor = \frac{1}{n}\sum_{i=1}^{n} P_i \tag{8}$$

The average points conceded per game are calculated as:

$$AvgPointsAgainst = \frac{1}{n}\sum_{i=1}^{n} O_i \tag{9}$$

The average point differential is defined as:

$$AvgPointDiff = AvgPointsFor - AvgPointsAgainst \tag{10}$$

For each match i, the point differential is defined as:

$$Diff_i = P_i - O_i \tag{11}$$

The mean point differential across all matches is computed as:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} Diff_i \tag{12}$$

Game outcomes are classified as:

$$Outcome_i = \begin{cases} Win, & P_i > O_i \\ Loss, & P_i < O_i \\ Draw, & P_i = O_i \end{cases} \tag{13}$$

The win percentage is calculated as:

$$Win\% = \frac{Wins}{TotalGames} \times 100 \tag{14}$$

where

$$TotalGames = Wins + Losses + Draws \tag{15}$$

To evaluate the consistency of team performance, the standard deviation of point differentials is calculated as:

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Diff_i - \mu)^2} \tag{16}$$

Finally, a team stability score is derived as:

$$Stability = \max\left(0, \min\left(100, 100 - 3\sigma\right)\right) \tag{17}$$

Through these analytical computations, the system converts raw game statistics into meaningful insights that support athlete evaluation, team performance assessment, and long-term sports analytics. The resulting metrics are visualised through dashboards and graphical summaries, enabling athletes and coaches to interpret performance trends and make data-driven decisions.

E. Athlete Portfolio Generation

The system also supports the creation of digital athlete portfolios by aggregating performance statistics, game history, achievements, and activity updates within individual athlete profiles. These portfolios provide a structured representation of an athlete's sporting history and performance metrics. By consolidating performance data into a unified profile, the system allows athletes to maintain comprehensive documentation of their athletic progress over time.

Overall, the proposed architecture enables centralized data collection, structured storage of performance records, and analytical evaluation of athlete performance, thereby supporting more effective documentation and analysis of sports activities.

IV. Database Design

The proposed system utilizes a relational database architecture to manage structured sports data related to athletes, teams, games, and player performance statistics. The schema is designed using a normalized relational model to ensure data integrity, maintain clear entity relationships, and support efficient querying for performance analytics.

The database separates core user identity, role-specific profiles, team participation, match records, and player statistics, enabling scalable management of sports data while supporting analytical operations such as performance summaries and dashboards.

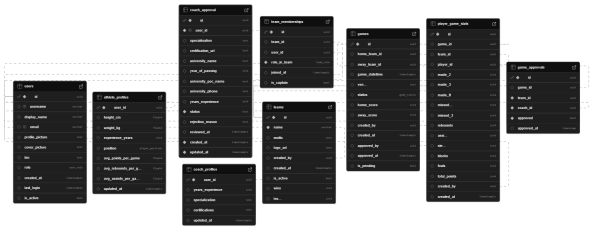


Fig. 2. Relational Database Schema for the Proposed Sports Data Management System

A. Core Entities

The primary entities in the database include *Users*, *Athlete Profiles*, *Coach Profiles*, *Teams*, *Team Memberships*, *Games*, and *Player Game Statistics*.

- **Users** serves as the central identity table, storing common user attributes such as username, email, display name, and role.
- **Athlete Profiles** and **Coach Profiles** extend the user entity with role-specific attributes such as playing position, experience, and specialization.

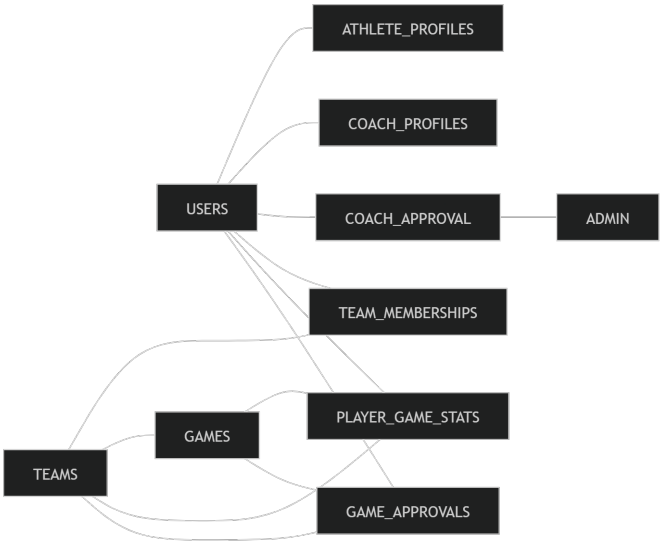


Fig. 3. High-level entity relationship diagram of the Rebound database architecture.

- **Teams** represents sports teams within the system.
- **Team Memberships** models the many-to-many relationship between users and teams.
- **Games** stores match-level information including participating teams, match schedule, and final scores.
- **Player Game Statistics** records individual athlete performance for a specific game.

B. Entity Relationships

The database structure establishes relationships that support performance tracking across multiple games.

- A user may have one athlete or coach profile depending on the assigned role.
- Users and teams share a many-to-many relationship through the *team_memberships* table.
- A game involves two teams and may contain multiple player statistics records.
- An athlete may have multiple performance records across different games.

These relationships allow the system to maintain complete match histories and track athlete performance over time.

C. Analytical Data Support

To support performance analysis and dashboard operations, the system uses database views and materialized views that aggregate player and team statistics. These structures enable efficient retrieval of metrics such as average player performance and team-level summaries without impacting transactional operations.

Foreign key constraints and unique indexes are used to ensure referential integrity, prevent duplicate records, and maintain consistency across game logs and player statistics. This database design enables reliable storage and efficient analysis of sports performance data within the centralized game logging system.

V. Results and Analysis

The proposed centralized game logging system was evaluated by recording and analyzing basketball game statistics within the platform. The collected data was used to assess the system's ability to maintain structured performance records and generate analytical insights from accumulated game logs.

A. Structured Game Documentation

The implementation of centralized game logs enabled the systematic recording of match information, including participating teams, match outcomes, and individual player statistics. Each recorded game generated a structured dataset linking athletes, teams, and statistical contributions within a single match record. This structure allowed the system to maintain a complete match history for athletes across multiple games.

The centralized storage of game logs improved the consistency of performance records compared to informal tracking methods, ensuring that statistical data remained organized and accessible for analysis.

B. Athlete Performance Analysis

Using the recorded player statistics, the system generated analytical metrics that summarized individual athlete performance over time. Key performance indicators such as **average points per game, rebounds, assists, and shooting performance** were computed from aggregated game statistics.

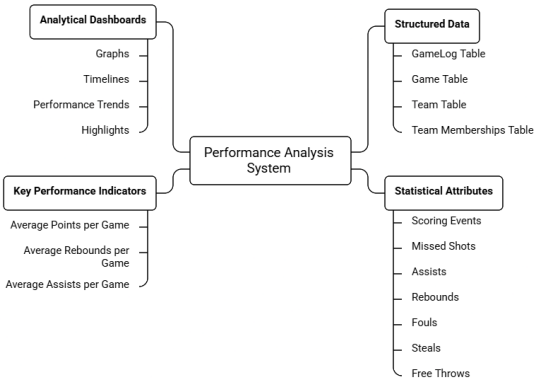


Fig. 4. Athlete Performance Analysis System

These metrics were visualized through analytical dashboards that displayed performance trends across multiple matches. The graphical representation of statistical data allowed users to identify patterns such as improvements in scoring efficiency or consistency in specific performance metrics.

C. Team-Level Insights

In addition to individual analysis, the system aggregated player statistics to generate team-level insights. The analysis enabled the identification of top-performing players within teams, most valuable players in specific matches, and standout individual game performances. These insights provided additional context to individual performance metrics and contributed to a broader understanding of team dynamics.

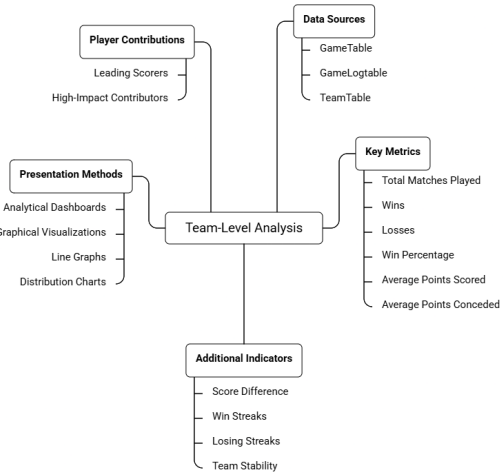


Fig. 5. Team-Level Analysis in Sports Analytics

D. System Effectiveness

The results demonstrate that centralized game logging enables efficient organization of performance data and supports analytical evaluation of athlete contributions. By maintaining structured datasets and generating automated performance metrics, the system simplifies the process of tracking athlete development across multiple games.

Overall, the evaluation indicates that centralized sports data management can significantly improve performance documentation and enable meaningful analysis for athletes and coaches operating in amateur sports environments.

VI. Discussion

The results demonstrate that centralized game logging enables structured documentation of sports performance and improves the accessibility of athlete data. By organizing game records and player statistics within a unified system, athletes and coaches can maintain consistent performance histories and analyze trends across multiple games. The integration of performance analytics further enhances the usefulness of the system by transforming raw statistics into interpretable insights through visual summaries and aggregated metrics.

However, the system relies on manually entered statistics, which may introduce inaccuracies if data is recorded or monitored incorrectly or incompletely. Additionally, the current implementation focuses on structured statistical data and does not incorporate automated tracking technologies. Future work may explore the integration of automated data capture methods and advanced analytics to further enhance performance evaluation capabilities.

VII. Conclusion

This study presented the design and evaluation of a centralized game logging system developed to support athlete performance analysis. The proposed platform enables athletes and coaches to record game data, track player statistics, and generate analytical insights through a structured and centralized data management framework. By organizing performance

data across multiple games, the system facilitates systematic tracking of athlete development and match history.

The findings indicate that centralized game logging improves the accessibility and organization of sports performance data while enabling meaningful analytical insights. Such systems can support improved documentation and evaluation of athlete performance in amateur sports environments, providing a foundation for more data-driven performance analysis and athlete development.

VIII. Future Work

While the proposed system demonstrates the effectiveness of centralized game logging for athlete performance analysis, several enhancements can further extend its capabilities. Future work may explore the integration of automated data collection technologies, such as computer vision–based tracking or wearable sensors, to reduce reliance on manual data entry and improve the accuracy of recorded statistics.

Additionally, the system can be expanded to support advanced analytics and predictive models that analyze long-term performance trends and provide recommendations for athlete development. Extending the platform to support multiple sports and larger datasets would also improve its applicability across different athletic environments and enable broader performance comparisons at both the individual and team levels.

References

[1] A. Ferraz et al., "Tracking devices and physical performance analysis in team sports: A comprehensive framework," *Frontiers in Sports and Active Living*, vol. 5, 2023.

[2] H. R. Thornton, J. A. Delaney, G. M. Duthie, and B. J. Dascombe, "Developing athlete monitoring systems in team sports: Data analysis and visualization," *International Journal of Sports Physiology and Performance*, vol. 14, no. 2, pp. 230–240, 2019. :contentReference[oaicite:1]index=1

[3] A. Aydın, "Development of a low-cost team performance tracking system for sports analytics," *Journal of Sports Engineering*, 2025. :contentReference[oaicite:2]index=2

[4] X. Tang, B. Long, and L. Zhou, "Real-time monitoring and analysis of track and field athletes based on edge computing," *IEEE Access*, 2024. :contentReference[oaicite:3]index=3

[5] V. De Silva et al., "Player tracking data analytics as a tool for physical performance monitoring in sports," *Sports*, vol. 6, no. 4, pp. 130–145, 2018. :contentReference[oaicite:4]index=4

[6] Y. Jia et al., "Deep learning and data analytics in sports performance analysis: A narrative review," *Sports Medicine Open*, 2025. :contentReference[oaicite:5]index=5

[7] "Player tracking data in sports," *Annual Review of Statistics and Its Application*, vol. 10, pp. 1–24, 2023. :contentReference[oaicite:6]index=6

[8] J. L. Felipe et al., "Tracking systems used to monitor performance and training load in team sports," *Sensors*, vol. 21, no. 24, 2021. :contentReference[oaicite:7]index=7

[9] Q. Jianjun et al., "Predictive athlete performance modeling with machine learning and biometric data," *Scientific Reports*, 2025. :contentReference[oaicite:8]index=8

[10] X. Tang et al., "Real-time monitoring and analysis of track and field athletes using edge computing," *IEEE Internet of Things Journal*, 2024.

[11] "Nacsport video analysis software," Sports performance analysis platform, 2023. :contentReference[oaicite:9]index=9

[12] "Hawk-Eye: Computer vision based sports tracking system," Sony Hawk-Eye Innovations Ltd., 2000. :contentReference[oaicite:10]index=10

[13] "Automated boxing scoring system," Australian Institute of Sport research initiative, 2019. :contentReference[oaicite:11]index=11

[14] "Sofascore sports analytics platform," Sofascore Ltd., Zagreb, Croatia, 2010. :contentReference[oaicite:12]index=12

[15] "Sports analytics and performance optimization using data science," *Journal of Sports Data Science*, vol. 4, no. 2, pp. 85–102, 2024. :contentReference[oaicite:13]index=13